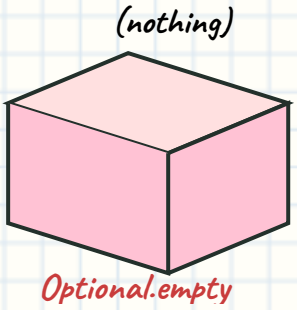
Optional · my notes LO5

a box that holds 0 or 1 value · the return type says "maybe nothing"



Optional[Alice Tan]



Optional.empty

`Optional.of(x)` x must not be null
`Optional.ofNullable(x)` null → empty
`Optional.empty()`
`Optional.of(null)` → NPE now!

✓ USE for

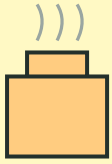
method **return types** where "no result" is normal:
`findCustomer` , `findPolicy` , `latestTransaction`

× NOT for

fields · parameters · `Optional<List>` · replacing every null

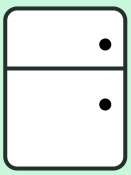
orElse = cook the backup meal ALWAYS · orElseGet = only if the fridge is empty

orElse(defaultName())



evaluated FIRST,
even for C001 (Alice)
→ wasted call

orElseGet(() -> defaultName())



runs ONLY when
the box is empty (C999)
→ lazy ✓

constant default → `orElse("UNKNOWN")` is fine · database / service / log / new object → `orElseGet`

Transform without opening the box

map



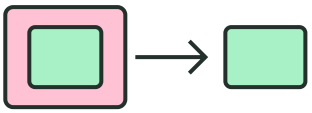
empty stays empty

filter



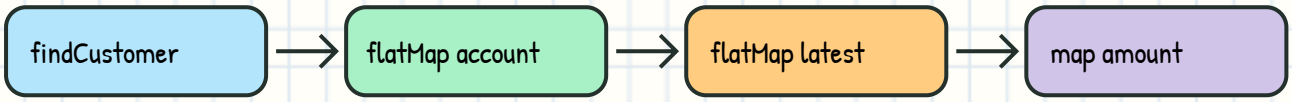
POL-509: 11-07 ✓ 11-13 ×

flatMap



box in a box → one box

The relay: one empty runner stops the race



C001 → Optional[447.37] · C003 FD → no txns · C005 CURRENT → no account · C999 → no customer

Toolbox

- `ifPresentOrElse` , `or` , `stream` [9]
- `orElseThrow()` [10] = honest `get()`
- `isEmpty()` [11]
- `flatMap(Optional::stream)` drops empties → [POL-502, POL-509]

× Don'ts

- `get()` → "No value present"
- `isPresent()` + `get()`
- `return null;` from Optional method
- Optional fields / params / lists

★ Remember

box for return types · orElse eager, orElseGet lazy · flatMap for lookups · empty list, not Optional<List> · Module 1 done! next → L06 Stream fundamentals